



## SYDNEY GIRLS HIGH SCHOOL

**2013**

TRIAL HIGHER SCHOOL CERTIFICATE EXAMINATION

# Mathematics Extension 2

### General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Board approved calculators may be used.
- A table of standard integrals is provided at the back of this paper.
- Diagrams are NOT drawn to scale.
- All necessary working should be shown in every question.
- Start each question on a new page.

**Total marks – 100**

### **Section I Pages 3-7**

**10 marks**

- Attempt Questions 1-10
- Allow about 15 minutes for this section

### **Section II Pages 8-17**

**90 marks**

- Attempt Questions 11-16
- Allow about 2 hours and 45 minutes for this section

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## Section I

10 marks

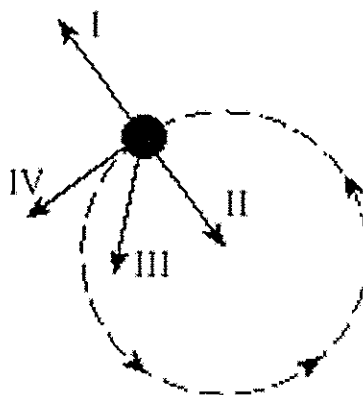
Attempt Questions 1–10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1–10.

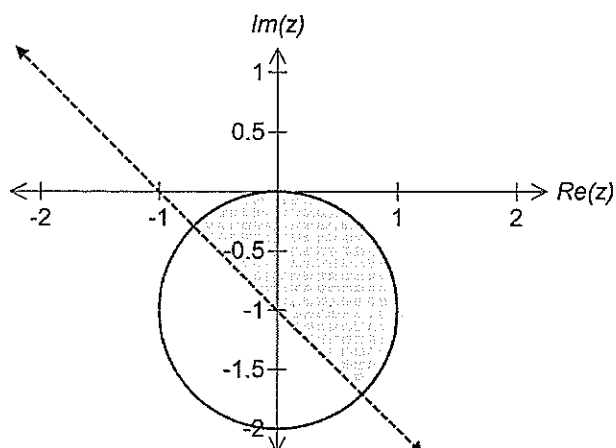
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- 1 An object moves in a circular path at a constant speed. Which vector in the diagram below best represents the object's acceleration?



- (A) I
- (B) II
- (C) III
- (D) IV
2. Which of the following cannot be the argument of a complex number  $z$  such that  $z^9 = -1 + i$ ?
- (A)  $\frac{11\pi}{36}$
- (B)  $\frac{\pi}{12}$
- (C)  $\frac{29\pi}{36}$
- (D)  $\frac{19\pi}{36}$

3. Consider the Argand diagram below:



Which pair of inequalities correctly define the shaded area?

- (A)  $|z+i| \leq 1$  and  $0 \leq \arg(z+1) < -\frac{\pi}{4}$
- (B)  $|z-i| \leq 1$  and  $0 \leq \arg(z-1) < -\frac{\pi}{4}$
- (C)  $|z-i| \leq 1$  and  $0 \leq \arg(z-1) < \frac{\pi}{4}$
- (D)  $|z+i| \leq 1$  and  $0 \leq \arg(z+1) < \frac{\pi}{4}$
4. A point  $P(x, y)$  moves so that the ratio of its distance from the point  $(1, 0)$  and the line  $x = 2$  is a constant  $e$ , where  $0 < e < 1$ . The locus described by this point would be a:
- (A) Circle
- (B) Parabola
- (C) Hyperbola
- (D) Ellipse.

5. Given that  $(x-1)P(x) = 16x^5 - 20x^3 + 5x - 1$ , then if  $P(x) = (4x^2 + ax - 1)^2$ , the value of  $a$  is:

- (A) 1
- (B) 2
- (C)  $\frac{1}{2}$
- (D) 0.

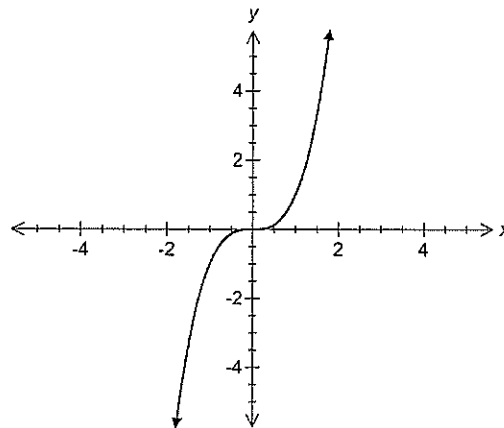
6.  $\int_0^{\pi} 5 \sin x \cos^4 x \, dx =$

- (A) 0
- (B) 2
- (C) -2
- (D) 20

7. Given that  $\int \sec^n x \, dx = \frac{1}{n-1} \tan x \sec^{n-2} x + \frac{n-2}{n-1} \int \sec^{n-2} x \, dx$  then  $\int_0^{\frac{\pi}{4}} \sec^4 x \, dx =$

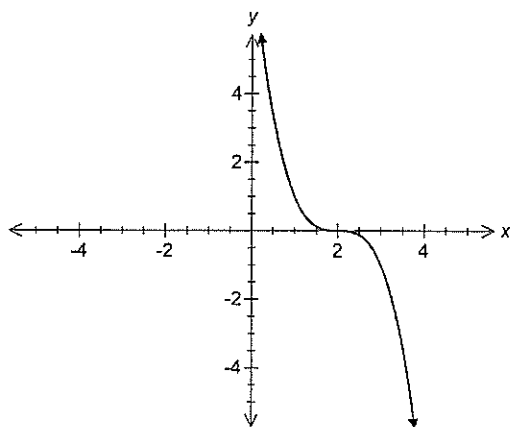
- (A)  $\frac{4}{3}$
- (B) 1
- (C)  $\frac{5}{6}$
- (D)  $\frac{6+4\sqrt{2}}{9}$

8. The graph of  $y = f(x)$  is shown below.

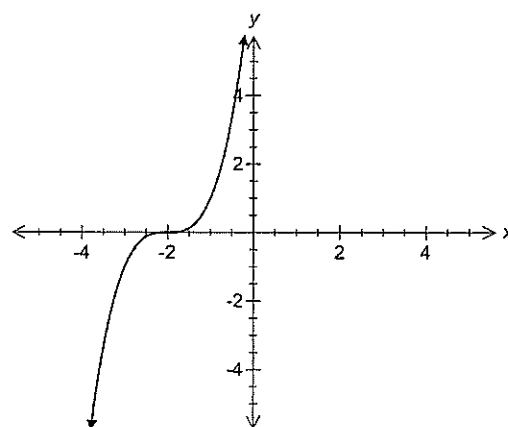


Which graph best represents  $y = f(2 - x)$ ?

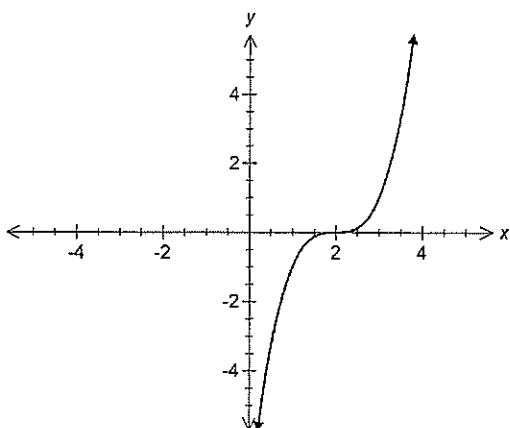
(A)



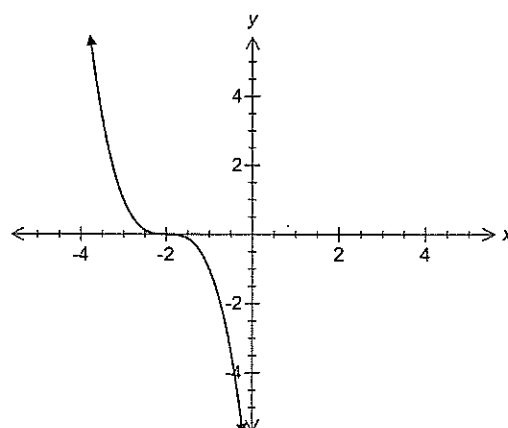
(B)



(C)



(D)



9. The curve  $4y^2 - 9x^2 = 36$  is defined parametrically by the equations:

(A)  $x = 4 \sec \theta, y = 9 \tan \theta$

(B)  $x = 3 \sec \theta, y = 2 \tan \theta$

(C)  $x = 2 \tan \theta, y = 3 \sec \theta$

(D)  $x = 9 \tan \theta, y = 4 \sec \theta$

10.  $\frac{3}{(x^2+2)(x-1)} = \frac{Px+Q}{x^2+2} + \frac{R}{x-1}$  where  $x \neq 1$ , and  $P, Q, R$  are constants. Which one of the following statements is false?

(A)  $3 = (Px+Q)(x-1) + R(x^2+2)$

(B)  $R = 1$

(C)  $3 = 2R - Q$

(D)  $3 = P + R$

## Section II

90 marks

Attempt Questions 11–16

Allow about 2 hours and 45 minutes for this section

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

In Questions 11–16, your responses should include relevant mathematical reasoning and/or calculations.

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### Question 11 (15 marks)

(a) The complex number  $w$  is given by  $w = -1 + \sqrt{3}i$ .

(i) Show that  $w^2 = 2\bar{w}$ . 2

(ii) Evaluate  $|w|$  and  $\arg w$ . 2

(iii) Show that  $w$  is a root of the equation  $w^3 - 8 = 0$ . 1

(b) Sketch the region of the Argand diagram whose points satisfy the inequalities

$|z - \bar{z}| \leq 4$  and  $\frac{-\pi}{3} \leq \arg z \leq \frac{\pi}{3}$ . 2

Question 11 continues on page 9



- (c)  $ABCD$  is a quadrilateral whose diagonals  $AC$  and  $BD$  are equal and bisect each other at the origin.  $A$  is the complex number  $z$  and  $\angle AOB = 30^\circ$ .
- (i) Find the coordinates of  $B, C$ , and  $D$  in terms of  $z$ . 2
- (ii) What type of quadrilateral is  $ABCD$ ? (justify your answer) 1

- (d) The complex number  $z$  is a function of the real number  $r$  given by the rule 3

$$z = \frac{r-i}{r+i}, \quad 0 \leq r \leq 1.$$

Evaluate  $|z|$  and hence describe the locus of  $z$  as  $r$  varies from 0 to 1.

- (e) By completing the square, find  $\int \frac{dx}{x^2 + 4x - 1}$ . 2

**End of Question 11**

**Question 12** (15 marks)

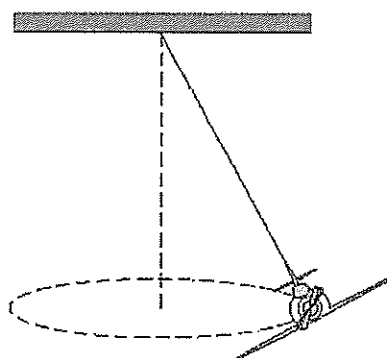
(a) (i) Express  $\frac{x+7}{x^2(x+2)}$  in the form  $\frac{A}{x} + \frac{B}{x^2} + \frac{C}{x+2}$ . 2

(ii) Hence or otherwise find  $\int \frac{x+7}{x^2(x+2)} dx$ . 2

(b) Find  $\int \frac{dx}{\sqrt{1+4x^2}}$ . 2

(c) Evaluate  $\int_0^1 xe^{-x} dx$ . 4

- (d) A toy airplane attached to a string 1.5m long moves in a horizontal circle as shown in the diagram below. The string makes an angle of  $30^\circ$  with the vertical. (Use  $g = 10 \text{ ms}^{-2}$ )



(i) What is the mass of the plane if the tension in the string is 5N? 3

(ii) Calculate the period of the plane's motion. 2

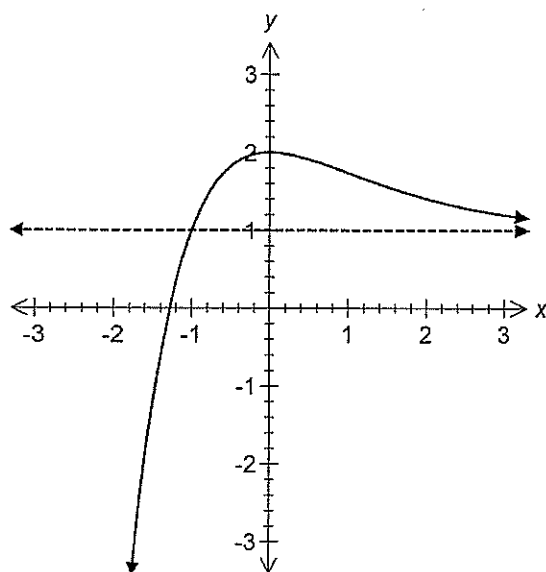
**End of Question 12**

**Question 13 (15 marks)**

(a) Using the substitution  $t = \tan\left(\frac{\theta}{2}\right)$ , show that  $\int_0^{\frac{\pi}{3}} \frac{1}{1 + \sin \theta} d\theta = \sqrt{3} - 1$  .

4

(b) The diagram shows the graph  $y = f(x)$  .



Draw separate one-third page sketches of the graphs of the following:

(i)  $y = f(|x|)$  1

(ii)  $y = \frac{1}{f(x)}$  2

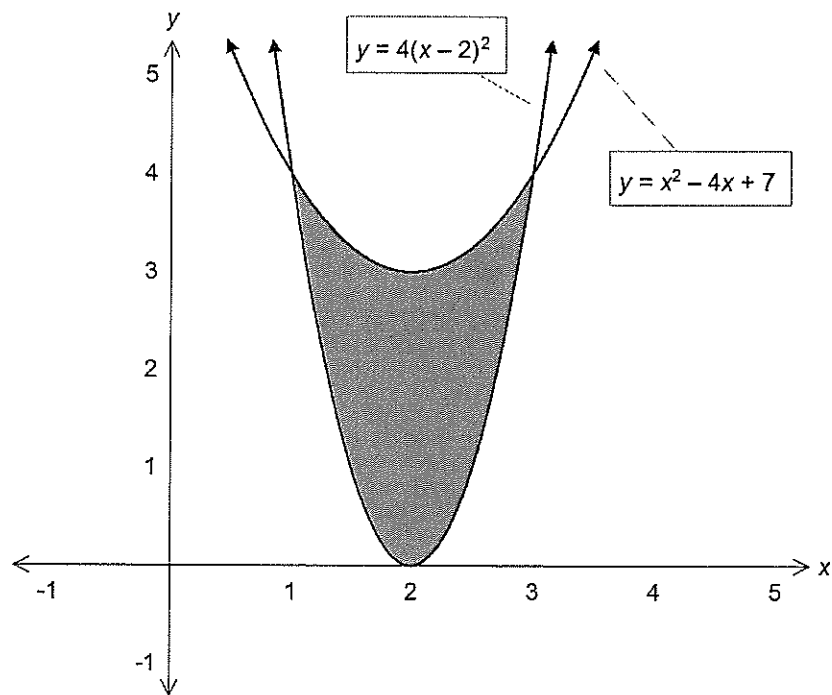
(iii)  $y = [f(x)]^2$  2

(iv)  $y = \ln[f(x)]$ . 2

**Question 13 continues on page 12**

- (c) Use the method of cylindrical shells to find the volume of the solid obtained by rotating the region between the curves  $y = 4(x-2)^2$  and  $y = x^2 - 4x + 7$  about the  $y$  axis.

4



**End of Question 13**

**Question 14 (15 marks)**

- (a) The Earth's orbit around the sun is elliptical with the Sun as one of the foci. The semi major axis of the Earth's orbit is  $1.486 \times 10^8$  kilometres and its eccentricity is 0.017.
- (i) How close to the sun does the earth come? 2
- (ii) What is the greatest possible distance between the sun and the earth? 2
- (b) Determine all the roots of the equation  $x^4 - 5x^3 - 9x^2 + 81x - 108 = 0$  given that there is a root of multiplicity 3. 3
- (c) The polynomial  $P(x) = x^3 + ax^2 + bx + 6$  where  $a, b$  are real numbers has  $1 - i$  as one zero.
- (i) Find  $a$  and  $b$ . 2
- (ii) Factorise  $P(x)$  over the complex field. 1
- (iii) Factorise  $P(x)$  over the real field. 1
- (d) Show that the equation  $3x^5 + 20x^3 + 45x = c$  can have only one real root, and find the value of the constant  $c$ , if the sum of the other (complex) roots is -7. 4

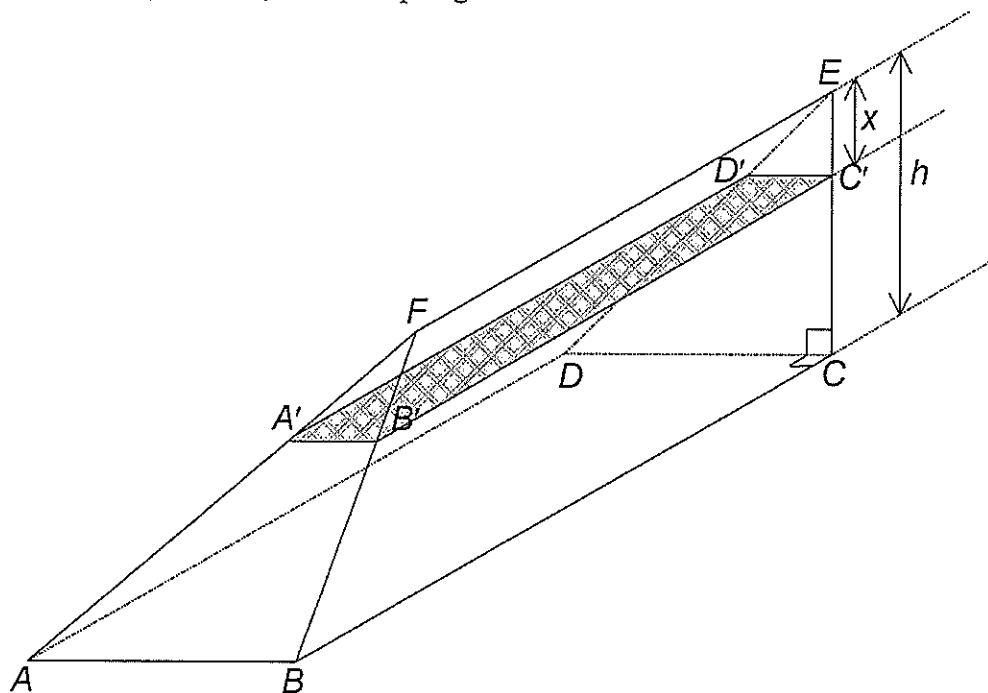
**End of Question 14**

**Question 15 (15 Marks)**

- (a) For the hyperbola  $\frac{x^2}{25} - \frac{y^2}{16} = 1$ , find:
- (i) its eccentricity 1
  - (ii) the coordinates of its foci 1
  - (iii) the equations of its directrices 1
  - (iv) the equations of its asymptotes 1
  - (v) the equation of the chord of contact of tangents drawn from the point (1,2) 1
- (b) Consider the region enclosed by the curves  $x = y^2$  and  $x = 2 - y^2$ . Find the volume of the solid formed when this region is rotated about the line  $x = 3$  by taking slices perpendicular to the axis of rotation. 4

**Question 15 continues on page 15**

- (c) Consider a solid  $ABCDEF$  whose height is  $h$ , and whose base is a rectangle  $ABCD$ , where  $AB = a$ ,  $BC = b$ , and the top edge  $EF = c$



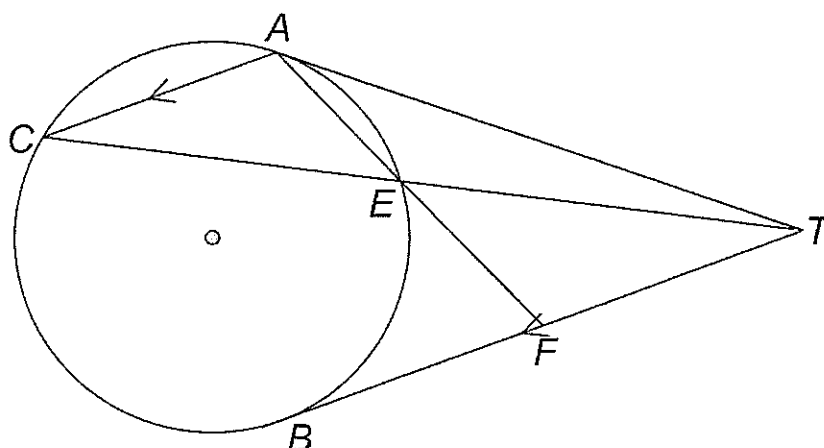
Consider a rectangular slice  $A'B'C'D'$  (parallel to the base  $ABCD$ )  $x$  units from the top edge, with width  $\Delta x$ .

- (i) Show that the volume of the slice is  $\Delta V = \left( \frac{x}{h} a \right) \left( c + \frac{b-c}{h} x \right) \Delta x$ . 3
- (ii) Hence show that the volume of the solid is  $\frac{ha}{6}(2b+c)$ . 3

**End of question 15**

**Question 16 (15 marks)**

- (a) Two tangents TA, TB are drawn from a point T to a given circle. Through A, a chord AC is drawn parallel to the other tangent TB and TC meets the circle at E.



- (i) Prove  $\triangle AFT$  is similar to  $\triangle EFT$ . 2
- (ii) Hence show that  $TF^2 = AF \times EF$ . 1
- (iii) Hence or otherwise prove that  $AE$  extended bisects  $TB$ . 2
- (b) (i) Prove that the equation of the tangent to the curve  $x = \frac{1}{1+t^4}$ ,  $y = \frac{t^5}{1+t^4}$  3  
at the point with parameter  $t$  is  $4y + (5t + t^5)x = 5t$ .
- (ii) This tangent meets the coordinate axes  $OX$  and  $OY$  in the points  $P$  and  $Q$ . 3  
Show that the area of the triangle  $OPQ$  never exceeds  $\left(\frac{15}{32}\right)\left(\frac{5}{3}\right)^{\frac{1}{4}}$ .

**Question 16 continues on page 17**



- (c) Prove by mathematical induction for all positive integers  $n$  ,

4

$$\tan^{-1}\left(\frac{1}{2 \times 1^2}\right) + \tan^{-1}\left(\frac{1}{2 \times 2^2}\right) + \dots + \tan^{-1}\left(\frac{1}{2 \times n^2}\right) = \frac{\pi}{4} - \tan^{-1}\left(\frac{1}{2n+1}\right) .$$

**End of paper**

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## STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left( x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left( x + \sqrt{x^2 + a^2} \right)$$

NOTE:  $\ln x = \log_e x$ ,  $x > 0$

# Extension 2 Trial HSC 2013

## Solutions.

### Multiple Choice.

- 1) B
- 2) C
- 3) All responses marked correct
- 4) D
- 5) B
- 6) B
- 7) ~~X~~ A
- 8) A
- 9) C
- 10) D

11)

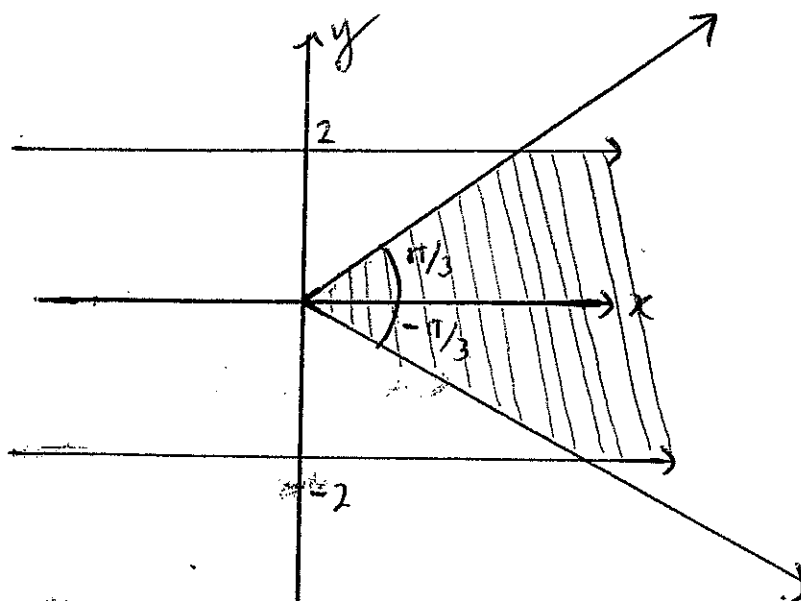
$$\begin{aligned}
 a) \quad i) \quad w^2 &= (-1 + \sqrt{3}i)^2 \\
 &= 1 - 2\sqrt{3}i - 3 \\
 &= -2 - 2\sqrt{3}i \\
 2\bar{w} &= 2(-1 - \sqrt{3}i) \\
 &= -2 - 2\sqrt{3}i \\
 &= w^2.
 \end{aligned}$$

$$\begin{aligned}
 ii) \quad |w| &= \sqrt{(-1)^2 + \sqrt{3}^2} \\
 &= 2
 \end{aligned}$$

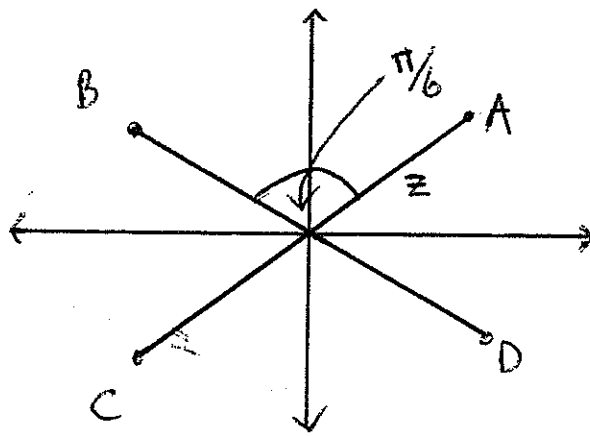
$$\arg w = 2\pi/3$$

$$\begin{aligned}
 iii) \quad w^3 - 8 &= [2 \operatorname{cis} 2\pi/3]^3 - 8 \\
 &= 8 \operatorname{cis} 2\pi - 8 \\
 &= 8 - 8 \\
 &= 0
 \end{aligned}$$

b)



c)



i)  $B = z \operatorname{cis}\left(\frac{\pi}{6}\right)$

$C = -z$

$D = -z \operatorname{cis}\left(\frac{\pi}{6}\right)$

ii) ABCD is a rectangle.  
(equal diagonals that bisect each other).

d)  $|z| = \frac{|r-i|}{|r+i|}$   
 $= \frac{\sqrt{r^2+1}}{\sqrt{r^2+1}}$   
 $= 1$

When  $r=0$   $z = \frac{-i}{i} = -1$

$r=1$   $z = \frac{1-i}{1+i} \times \frac{(1-i)}{(1-i)}$   
 $= -i$

$\therefore$  locus is a quarter of the circle  $x^2 + y^2 = 1$   
in the 3rd quadrant.

$$\begin{aligned} e) \int \frac{dx}{x^2 + 4x - 1} &= \int \frac{dx}{(x+2)^2 - 5} \\ &= \frac{1}{2\sqrt{5}} \left[ \ln \left( \frac{x+2-\sqrt{5}}{x+2+\sqrt{5}} \right) \right] + C \end{aligned}$$

$$12 (a)(i) \quad x+7 = Ax(x+2) + B(x+2) + Cx^2$$

$$\text{when } x=0$$

$$7 = 2B$$

$$B = \frac{7}{2}$$

$$\text{when } x=-2$$

$$5 = 4C$$

$$C = \frac{5}{4}$$

$$\text{when } x=1$$

$$8 = 3A + 3B + C$$

$$= 3A + \frac{21}{2} + \frac{5}{4}$$

$$A = -\frac{5}{4}$$

$$(ii) \int \left( -\frac{5}{4x} + \frac{7}{2x^2} + \frac{5}{4(x+2)} \right) dx$$

$$= -\frac{5}{4} \log x - \frac{7}{2x} + \frac{5}{4} \log(x+2) + c$$

$$(b) \frac{1}{2} \int \frac{dx}{\sqrt{\frac{1}{4} + x^2}} = \frac{1}{2} \log \left( x + \sqrt{\frac{1}{4} + x^2} \right) + c$$

$$(c) \text{ Let } u = x \quad v' = e^{-x}$$

$$u' = 1 \quad v = -e^{-x}$$

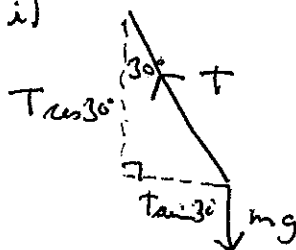
$$\int_0^1 x e^{-x} dx = [-x e^{-x}]_0^1 + \int_0^1 e^{-x} dx$$

$$= -e^{-1} + 0 + [-e^{-x}]_0^1$$

$$= -e^{-1} - e^{-1} + 1$$

$$= 1 - 2e^{-1}$$

(d) i)



$$T \cos 60^\circ = mg$$

$$5 \cos 30^\circ = m \times 10$$

$$m = \frac{1}{2} \times \frac{\sqrt{3}}{2}$$

$$= \frac{\sqrt{3}}{4}$$

ii)

$$T = m l \omega^2$$

$$5 = \frac{\sqrt{3}}{4} \times \frac{3}{2} \times \omega^2$$

$$\frac{40}{3\sqrt{3}} = \omega^2$$

$$\omega = \sqrt{\frac{40}{3\sqrt{3}}}$$

$$\text{Period} = \frac{2\pi}{\omega} = \frac{2\pi}{\sqrt{\frac{40}{3\sqrt{3}}}}$$

$$= 2.264 \text{ sec} \approx 2.3 \text{ sec}$$



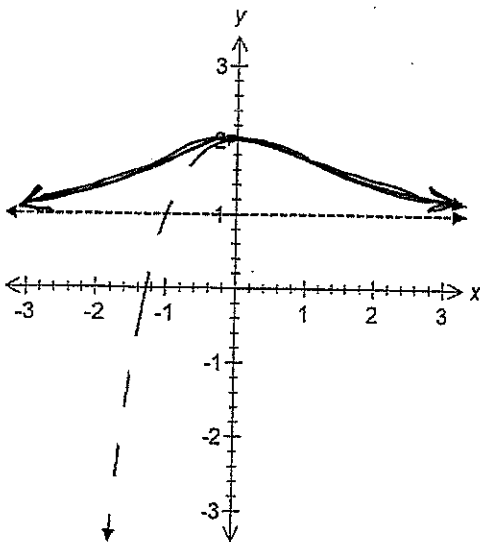
Question 13:

a)

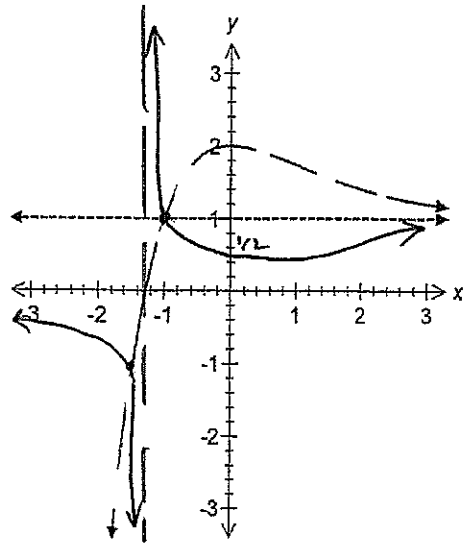
$$\begin{aligned} \int_0^{\frac{\pi}{3}} \frac{1}{1+\sin \theta} d\theta &= \int_0^{\frac{1}{\sqrt{3}}} \frac{1}{\left(1+\frac{2t}{1+t^2}\right)} \cdot \frac{2dt}{1+t^2} \\ &= \int_0^{\frac{1}{\sqrt{3}}} \frac{2dt}{1+t^2+2t} \\ &= \int_0^{\frac{1}{\sqrt{3}}} 2(t+1)^{-2} dt \\ &= \left[ \frac{2}{t+1} \right]_{\frac{1}{\sqrt{3}}}^0 \\ &= 2 - \frac{2\sqrt{3}}{1+\sqrt{3}} \\ &= \frac{2+2\sqrt{3}-2\sqrt{3}}{1+\sqrt{3}} \\ &= \frac{2}{1+\sqrt{3}} \times \frac{1-\sqrt{3}}{1-\sqrt{3}} \\ &= \frac{2(1-\sqrt{3})}{-2} \\ &= \sqrt{3}-1 \end{aligned}$$

b)

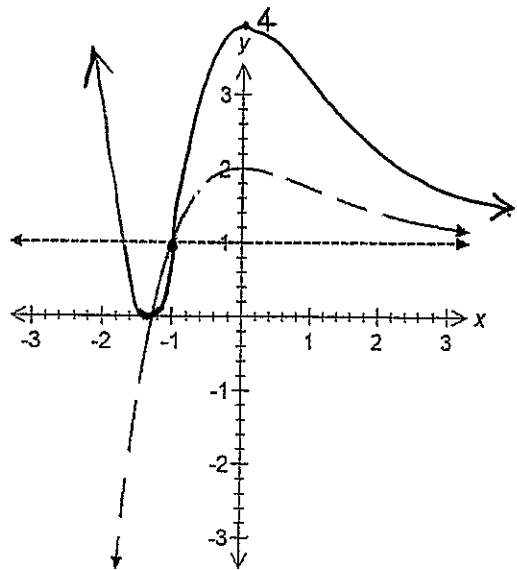
i)  $y = f(|x|)$



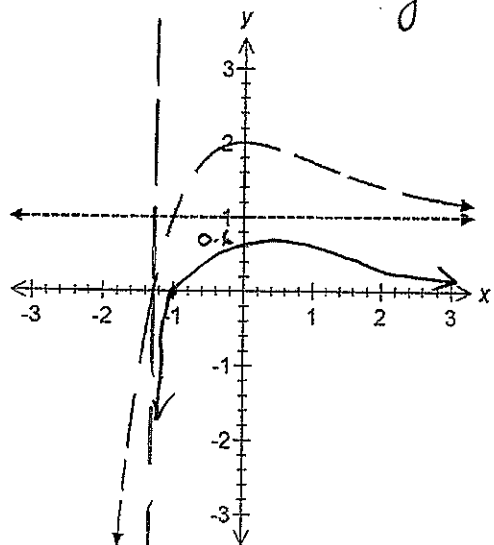
ii)  $y = \frac{1}{f(x)}$



iii)  $y = [f(x)]^2$

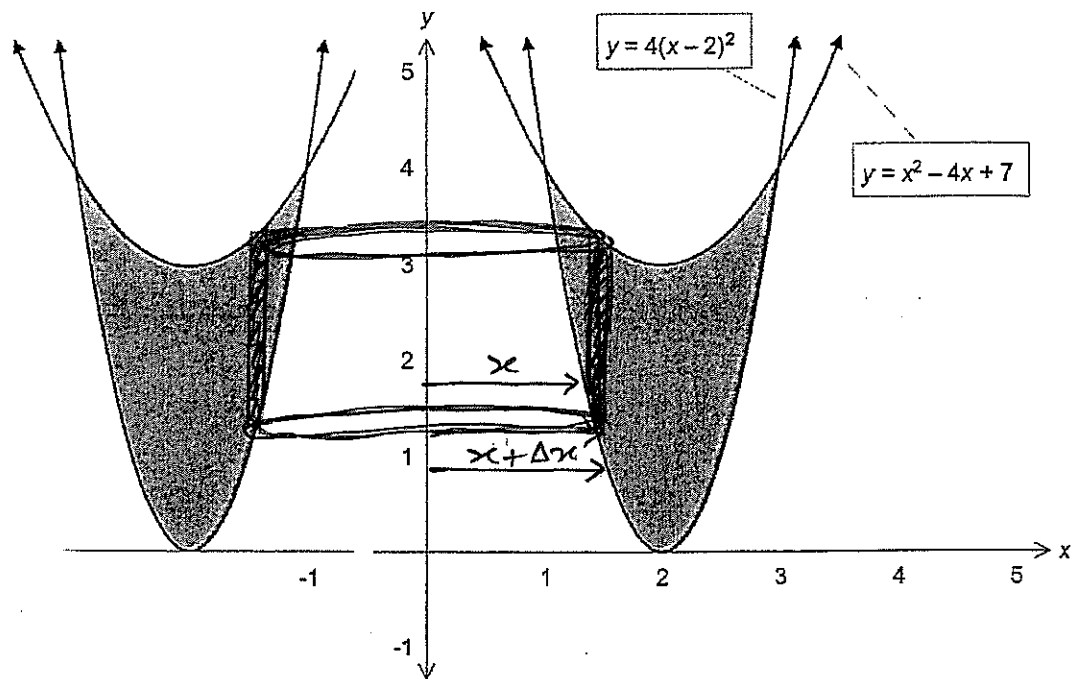


iv)  $y = \ln[f(x)]$



$y = \ln 2 \approx 0.6$

c)



Area of base of cylindrical shell:  $\Delta A(x) = \pi(R^2 - r^2)$

$$\Delta A(x) = \pi((x + \Delta x)^2 - r^2)$$

$$\Delta A(x) = 2\pi x \Delta x, \text{ (since } \Delta x^2 \text{ is very small).}$$

Volume of shell:

$$\Delta V(x) = \Delta A(x) \times \text{height}$$

$$\Delta V(x) = \Delta A(x) \times (x^2 - 4x + 7 - 4(x-2)^2)$$

$$\Delta V(x) = \Delta A(x) \times (-3x^2 + 12x - 9)$$

$$\Delta V(x) = 2\pi x \times 3(-x^2 + 4x - 3) \Delta x$$

Volume of solid:

$$V = \lim_{\Delta x \rightarrow 0} \sum_{x=1}^3 6\pi(-x^3 + 4x^2 - 3x) \Delta x$$

$$= 6\pi \left[ -\frac{x^4}{4} + \frac{4x^3}{3} - \frac{3x^2}{2} \right]_1^3$$

$$= 6\pi \left( \left( -\frac{81}{4} + \frac{108}{3} - \frac{27}{2} \right) - \left( -\frac{1}{4} + \frac{4}{3} - \frac{3}{2} \right) \right)$$

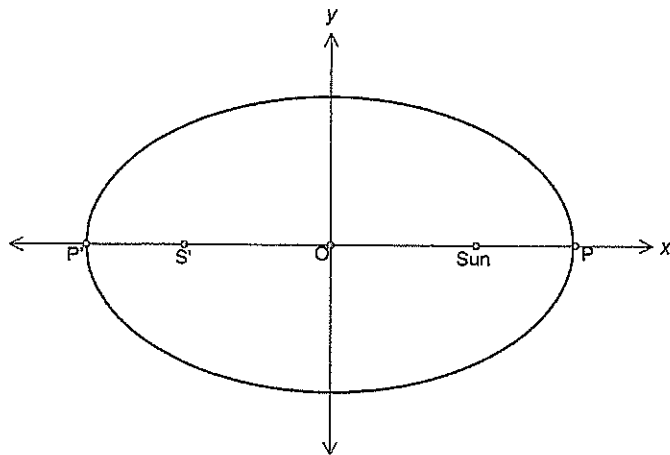
$$= 6\pi \left( \frac{9}{4} - \left( -\frac{5}{12} \right) \right)$$

$$= 6\pi \times \frac{8}{3}$$

$$\therefore V = 16\pi \text{ units}^3$$

Question 14:

a)



i) Earth is closest to the Sun at P.

$$PS = PO - ae$$

$$= 1.486 \times 10^8 - (1.486 \times 10^8 \times 0.017)$$

$$= 1.486 \times 10^8 - 2526200$$

$$= 146073800 \text{ km}$$

$$= 1.460738 \times 10^8 \text{ km}$$

ii) Earth is furthest from the Sun at P'.

$$P'S = PO + ae$$

$$= 1.486 \times 10^8 + (1.486 \times 10^8 \times 0.017)$$

$$= 1.017 \times 1.486 \times 10^8$$

$$= 151126200 \text{ km}$$

$$= 1.511262 \times 10^8 \text{ km}$$

b)

Let

$$P(x) = x^4 - 5x^3 - 9x^2 + 81x - 108$$

$$P'(x) = 4x^3 - 15x^2 - 18x$$

$$P''(x) = 12x^2 - 30x - 18$$

For a root of multiplicity 3,  $P''(x) = 0$ .

That is:

$$0 = 12x^2 - 30x - 18$$

$$0 = 2x^2 - 5x - 3$$

$$0 = (x-3)(2x+1)$$

$$\therefore x = 3 \text{ or } x = -\frac{1}{2}$$

Testing roots:  $P(3) = 0$

$\therefore x = 3$  is a triple root.

Consider the sum of the roots of  $P(x)$ :

$$3 + 3 + 3 + \alpha = 5$$

$$\alpha = -4$$

$$\therefore P(x) = (x-3)^3(x+4) \text{ and}$$

$P(x)$  has roots 3, 3, 3 and -4.

c)

$$i) P(x) = x^3 + ax^2 + bx + 6$$

Since all the coefficients are real, then by the conjugate root theorem both  $(1-i)$  and  $(1+i)$  are factors of  $P(x)$ .

Product of roots of  $P(x)$ :

$$(1-i)(1+i)\alpha = -6$$

$$(1-i^2)\alpha = -6$$

$$2\alpha = -6$$

$$\alpha = -3$$

Now:

$$P(-3) = 0$$

$$0 = (-3)^3 + a(-3)^2 + b(-3) + 6$$

$$0 = -27 + 9a - 3b + 6$$

$$0 = -21 + 9a - 3b \dots \dots \dots (1)$$

Sum of roots:

$$(1-i) + (1-i) - 3 = -a$$

$$2-3 = -a$$

$$a = 1 \dots \dots \text{sub into (1)}$$

$$0 = -21 + 9(1) - 3b$$

$$0 = -12 - 3b$$

$$3b = -12$$

$$b = -4$$

ii)

Over the complex field:

$$P(x) = (x-1+i)(x-1-i)(x+3)$$

iii)

Over the real field:

$$\begin{aligned} P(x) &= ((x-1)^2 - i^2)(x+3) \\ &= (x^2 - 2x + 1 + 1)(x+3) \\ &= (x^2 - 2x + 2)(x+3) \end{aligned}$$

d)

Let

$$P(x) = 3x^5 + 20x^3 + 45x - c$$

$$P'(x) = 15x^4 + 60x^2 + 45$$

For stationary points  $P'(x) = 0$ .

$$0 = 15x^4 + 60x^2 + 45$$

$$0 = x^4 + 4x^2 + 3$$

$$0 = (x^2 + 3)(x^2 + 1)$$

No real solutions

No turning points.

For points of inflexion  $P''(x) = 0$ .

$$P''(x) = 60x^3 + 120x$$

$$0 = x(x^2 + 2)$$

$\therefore$  at  $x = 0$  ( $y = c$ ) there is a point of inflexion.

Hence there must be only one real root and two pairs of conjugate complex roots of  $P(x)$ .

Sum of 'other' roots is given as -7.

Consider the sum of the roots of  $P(x)$ :

$$-7 + \alpha = 0$$

$$\alpha = 7 \text{ (is the real root)}$$

Now  $P(7) = 0$  and by substitution  $c = 57596$ .

$$15 (a) (i) 16 = 25(e^2 - 1)$$

$$\frac{16}{25} = e^2 - 1$$

$$\frac{41}{25} = e^2$$

$$e = \frac{\sqrt{41}}{5} \checkmark$$

$$(ii) \left( \pm 5 \times \frac{\sqrt{41}}{5}, 0 \right)$$

$$= (\pm \sqrt{41}, 0) \checkmark$$

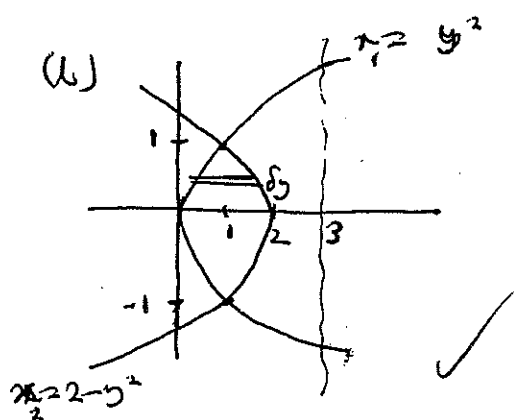
$$(iii) y = \pm \frac{5}{\sqrt{41}}$$

$$= \pm \frac{25}{\sqrt{41}} \checkmark$$

$$(iv) y = \pm \frac{4}{5}x \checkmark$$

$$(v) \frac{1 \times x}{25} - \frac{2 \times y}{16} = 1$$

$$\frac{x}{25} - \frac{y}{8} = 1 \checkmark$$



$$V_{shell} = \pi(r_2^2 - r_1^2)$$

$$= \pi \{ (3 - x_1)^2 - (3 - x_2)^2 \} dy$$

$$= \pi (9 - 6x_1 + x_1^2 - 9 + 6x_2 - x_2^2) dy$$

$$= \pi (6x_2 - 6x_1 + x_1^2 - x_2^2) dy$$

$$V_{solid} = \lim_{dy \rightarrow 0} \sum_{y=-1}^1 \pi (6x_2 - 6x_1 + x_1^2 - x_2^2) dy$$

$$= \pi \int_{-1}^1 (6x_2 - 6x_1 + x_1^2 - x_2^2) dy$$

$$= \pi \int_{-1}^1 (12 - 6y^2 - 6y^2 + y^4 - 9 + 9y^2 - y^4) dy$$

$$(c) V = Ah$$

$$= A'B' \cdot B'C' \cdot \delta x$$

$$A'B' = mx + k \checkmark$$

$$0 = m \cdot 0 + k$$

$$k = 0$$

$$\therefore A'B' = mx$$

$$h = mh$$

$$m = \frac{a}{h}$$

$$\therefore A'B' = \frac{a}{h}x$$

$$B'C' = mx + h$$

$$c = m \cdot 0 + h$$

$$h = c$$

$$B'C' = mx + c \checkmark$$

$$h = mh + c$$

$$m = \frac{h-c}{h}$$

$$\therefore B'C' = \frac{h-c}{h}x + c$$

$$\therefore V = \left( \frac{a}{h}x \right) \left( \frac{h-c}{h}x + c \right) \delta x$$

$$(ii) V = \lim_{h \rightarrow 0} \sum_{x=0}^h \left( \frac{a}{h}x \right) \left( c + \frac{h-c}{h}x \right) \delta x$$

$$= \frac{a}{h} \int_0^h x \left( c + \frac{h-c}{h}x \right) dx \checkmark$$

$$= \frac{a}{h} \int_0^h \left( cx + \frac{h-c}{h}x^2 \right) dx$$

$$= \frac{a}{h} \left[ \frac{cx^2}{2} + \frac{h-c}{2h}x^3 \right]_0^h \checkmark$$

$$= \frac{a}{h} \left( \frac{ch^2}{2} + \frac{h-c}{2h}h^3 \right) \checkmark$$

$$= \frac{ach}{2} + \frac{ah^2 - ac}{3}h$$

$$= \frac{ach}{2} + \frac{ah^3}{3} \checkmark$$

$$= \frac{ah}{c} (c + 2h)$$

G.E.D.

$$= \pi \int_{-1}^1 (8 + dy^2) dy$$

$$= 16\pi \int_0^1 (1 + y^2) dy$$

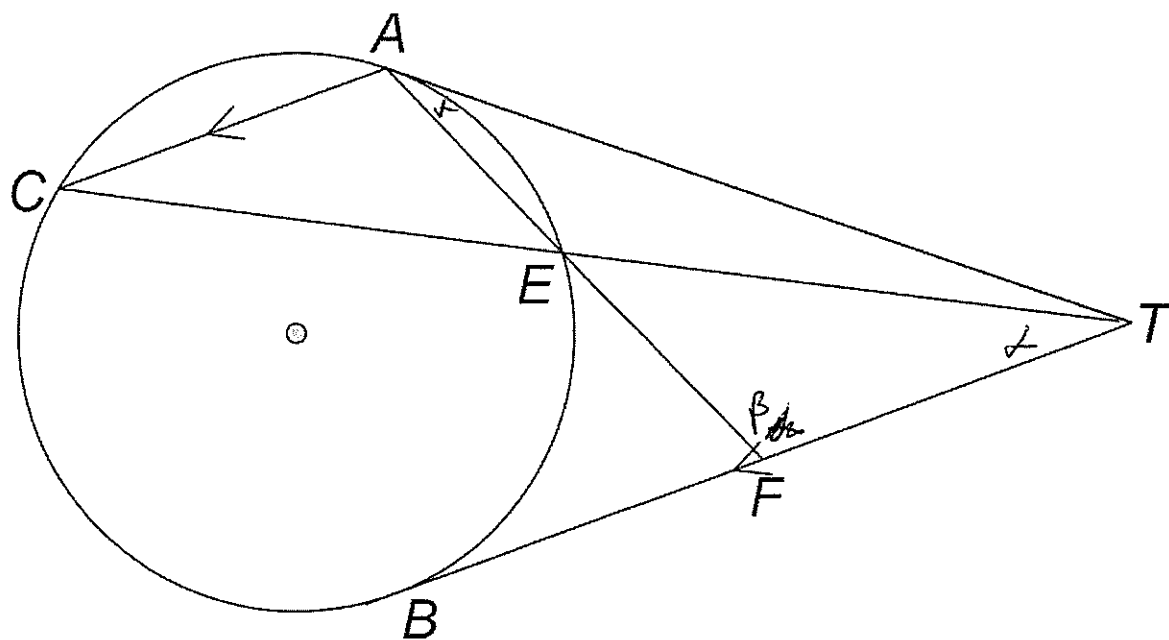
$$= 16\pi \left[ y + \frac{y^3}{3} \right]_0^1$$

$$= 16\pi \left( 1 + \frac{1}{3} \right) \checkmark$$

$$= \frac{32\pi}{3} u^3$$

Question 16.

a)



i) In  $\triangle TFA$  and  $\triangle ETF$

$$\angle TAF = \angle ACE = \alpha \text{ (}\angle \text{ in alt segment)}$$

$$\angle ACE = \angle ETF = \alpha \text{ (alt } \angle \text{ s } AC \parallel BT)$$

$$\therefore \angle TAF = \angle ETF$$

$$\angle EFT = \angle TFA \text{ (common)}$$

$$\therefore \triangle TFA \sim \triangle ETF \text{ (equiangular)}$$

$$\text{ii) } \frac{TF}{EF} = \frac{AF}{TF} \text{ (corresp sides of sim } \Delta \text{ s in same ratio)}$$

$$TF^2 = AF \times EF$$

$$\text{iii) } BF^2 = AF \times EF \text{ (tangent}^2 = \text{prod. of intercepts)}$$

$$\therefore TF^2 = BF^2$$

$$TF = BF$$

$\therefore$  AE extended bisects BT.

$$i) x = \frac{1}{1+t^4}$$

$$\frac{dx}{dt} = \frac{-4t^3}{(1+t^4)^2}$$

$$y = \frac{t^5}{1+t^4}$$

$$\frac{dy}{dt} = \frac{5t^4(1+t^4) - t^5(4t^3)}{(1+t^4)^2}$$

$$= \frac{5t^4 + 8t^8}{(1+t^4)^2}$$

$$= \frac{t^4(5+t^4)}{(1+t^4)^2}$$

$$\frac{dy}{dx} = \frac{dy}{dt} \cdot \frac{dt}{dx}$$

$$= \frac{t^4(5+t^4)}{(1+t^4)^2} \cdot \frac{(1+t^4)^2}{-4t^3}$$

$$= \frac{t(5+t^4)}{-4}$$

$$y - \frac{t^5}{1+t^4} = \frac{t(5+t^4)}{-4} \left[ x - \frac{1}{1+t^4} \right]$$

$$4y + x(5t + t^5) = \frac{4t^5}{1+t^4} + \frac{5t + t^5}{1+t^4}$$

$$= \frac{5t(1+t^4)}{(1+t^4)}$$

ii) When  $x=0$

$$4y = 5t$$

$$y = \frac{5t}{4}$$

$$\therefore OY = \frac{5t}{4}$$

When  $y=0$

$$x = \frac{5t}{5t + t^5}$$

$$= \frac{5}{5 + t^4}$$

$$\therefore OX = \frac{5}{5 + t^4}$$

$$\therefore A = \frac{1}{2} \times \frac{5t}{4} \times \frac{5}{5+t^4}$$

$$= \frac{25t}{8(5+t^4)}$$

$$\frac{dA}{dt} = \frac{25 \times 8(5+t^4) + 32t^3(25t)}{64(5+t^4)^2}$$

For a max area  $\frac{dA}{dt} = 0$

$$200(5+t^4) - 800t^4 = 0$$

$$1000 + 200t^4 - 800t^4 = 0$$

$$t^4 = \frac{5}{3}$$

$$t = \left(\frac{5}{3}\right)^{1/4}$$

$$\therefore \text{Max } A = \frac{1}{2} \times \frac{5}{4} \left(\frac{5}{3}\right)^{1/4} \times \frac{5}{5+5/3}$$

$$= \frac{5}{8} \left(\frac{5}{3}\right)^{1/4} \times \frac{3}{4}$$

$$= \frac{15}{32} \left(\frac{5}{3}\right)^{1/4}$$



c) Show true for  $n=1$

$$\text{LHS} = \tan^{-1}\left(\frac{1}{2}\right)$$

$$\begin{aligned}\text{RHS} &= \frac{\pi}{4} - \tan^{-1}\left(\frac{1}{3}\right) \\&= \tan^{-1}(1) - \tan^{-1}\left(\frac{1}{3}\right) \\&= \tan^{-1}\left(\frac{1 - \frac{1}{3}}{1 + \frac{1}{3}}\right) \\&= \tan^{-1}\left(\frac{1}{2}\right) \\&= \text{LHS}\end{aligned}$$

$\therefore$  true for  $n=1$

Assume true for  $n=k$

$$\tan^{-1}\left(\frac{1}{2 \times 1^2}\right) + \dots + \tan^{-1}\left(\frac{1}{2k^2}\right) = \frac{\pi}{4} - \tan^{-1}\left(\frac{1}{2k+1}\right)$$

Prove true for  $n=k+1$

R.T.P

$$\tan^{-1}\left(\frac{1}{2 \times 1^2}\right) + \dots + \tan^{-1}\left(\frac{1}{2(k+1)^2}\right) = \frac{\pi}{4} - \tan^{-1}\left(\frac{1}{2k+3}\right)$$

$$\text{LHS} = \frac{\pi}{4} - \tan^{-1}\left(\frac{1}{2k+1}\right) + \tan^{-1}\left(\frac{1}{2(k+1)^2}\right) \quad [\text{using assumption}]$$

$$= \frac{\pi}{4} + \tan^{-1}\left(\frac{1}{2(k+1)^2}\right) - \tan^{-1}\left(\frac{1}{2k+1}\right)$$

$$= \frac{\pi}{4} + \tan^{-1}\left[\left\{\frac{1}{2(k+1)^2} - \frac{1}{2k+1}\right\} \div \left\{1 + \frac{1}{2(k+1)^2} \times \frac{1}{2k+1}\right\}\right]$$

$$= \frac{\pi}{4} + \tan^{-1}\left[\frac{2k+1 - 2(k+1)^2}{2(2k+1)(k+1)^2} \div \frac{2(k+1)^2(2k+1) + 1}{2(2k+1)(k+1)^2}\right]$$

$$= \frac{\pi}{4} + \tan^{-1}\left[\frac{2k+1 - 2(k+1)^2}{2(k+1)^2(2k+1) + 1}\right]$$

$$= \frac{\pi}{4} + \tan^{-1} \left[ \frac{2k+1 - 2k^2 - 4k - 2}{(2k^2 + 4k + 2)(2k+1) + 1} \right]$$

$$= \frac{\pi}{4} + \tan^{-1} \left[ \frac{-2k^2 - 2k - 1}{4k^3 + 8k^2 + 4k + 2k^2 + 4k + 2 + 1} \right]$$

$$= \frac{\pi}{4} + \tan^{-1} \left[ \frac{-(2k^2 + k + 1)}{(2k+3)(2k^2 + k + 1)} \right]$$

$$= \frac{\pi}{4} + \tan^{-1} \left[ \frac{-1}{2k+3} \right]$$

$$= \frac{\pi}{4} - \tan^{-1} \left[ \frac{1}{2k+3} \right]$$

= RHS

∴ Proven true for  $n = k+1$ .

∴ Proposition is true for all  $n \geq 1$  by induction